Modelling the emergence of iconicity in interactive language learning

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TO DO

- Lapse rate?
- Edit overview figure
- Revisit the question of linear noise in a bounded system

Introduction

Iconicity refers to the linguistic phenomenon in which aspects of the form of a linguistic sign resembles aspects of its meaning (Winter, Woodin, and Perlman 2026).

For instance, referents that are small are commonly associated with different acoustic parameters than large references. Small referents tend to be associated with high front vowels (e.g. /i/), shorter duration, lower intensity, and higher pitch. Large referents on the other hand, are associated with low back vowels, longer duration, higher intensity, and lower pitch Winter and Perlman (2021). These sound-symbolic associations have been partly attributed to correlations between the physical size and in turn vocal tract dimensions of mammals and their resulting acoustic consequences (e.g. Fitch 1997; Ohala 1983).

In theoretical discussions of language, iconicity has often been regarded as playing only a marginal role. The conventionalized complexity of modern languages suggests that the relation between form and meaning is largely conventional or "arbitrary", and even if iconicity

could have been a driving force in the early emergence of language—facilitating the initial establishment of communicative conventions—it is expected to erode over time through semantic and phonological change (referred to as de-iconization by Flaksman 2017). Yet, modern languages display a plethora of iconic signs that are both sustained and possibly continuously reintroduced, indicating that iconicity coexist with other types of structure, such as systematicity and combinatoriality (Johansson, Anikin, and Aseyev 2020; Erben Johansson et al. 2020; Sidhu et al. 2021; Winter et al. 2022).

Empirically, the emergence of iconicity is often studied in experiments using artificial language learning. In these studies, an artificial language is represented by a set of form-meaning pairs, where the signal form is a written letter sequence, a spoken phoneme sequence, a gestured/signed manual movement sequence or other less-conventionalized continuous signal modulations (e.g. whistles, buzzer tones, etc.) (Hofer and Levy 2019; Kempe et al. 2019, 2021; Verhoef, Roberts, and Dingemanse 2015; Verhoef, Kirby, and De Boer 2016). In these experiments, the signal is assumed to convey a certain meaning such as a referent, often visually represented by images shown to participants. These images differ along perceptually salient semantic dimensions referring to e.g. shape (spiky/pointy vs. round/fluffy), motion (upward vs. downward movement), brightness (dark vs. light), or physical size (large vs. small). In this set-up, participants can thus use iconic aspects of the signal for the identification of referents or introduce and modulate iconicity while communicating.

In studies of iterative learning, participants learn the form-meaning mappings and reproduce them multiple times, with (as in language interaction games) or without an interlocutor. The language evolves across participant generations, where the reproductions of the first generation becomes the training input for the second, etc. Across trials, variability in pronunciation arises as participants tend to produce slightly different reproductions of the form. This variability can be affected by different biases that are for example rooted in cognitive constraints by the signal senders and receivers such as learning and memory constraints that favour some signals over others.

Arguably, iconicity is one possible result of these biases. For instance, a signal that resembles aspects of the intended meaning of a referent might be easier to identify, learn, memorize and/or reproduce.

Previous work has indeed found iconicity to emerge during artificial language learning under certain circumstances. Overall, in settings where there is communicative pressure for successful interaction, such as a language learning game between two people, participants have been shown to use and continuously reuse iconic strategies (Kempe et al. 2019; Verhoef, Roberts, and Dingemanse 2015). These strategies can be further reinforced by rewarding participants for successful interactions (earning points, stars, etc.).

However, as languages conventionalize, iconicity seems to loose its importance. Several studies have reported a trade-off between iconicity and the emergence of structured, combinato-

rial systems. For example, Hofer and Levy (2019) found that iconic mappings developed rapidly but gradually faded as combinatorial structure increased over successive interactions. Other work has shown that iconic forms can instead stabilize or even increase over repeated interactions (Perlman, Dale, and Lupyan 2015; Verhoef, Roberts, and Dingemanse 2015). More broadly, studies of artificial language evolution demonstrate that languages become more learnable over time—through reductions in complexity and/or increases in combinatorial structure—while iconicity either stabilizes or declines (Hofer and Levy 2019; Vinson et al. 2021). Verhoef and colleagues (2016) further observed that iconic forms persisted longer under conditions where opportunities for structural regularization were delayed, such as when form–meaning mappings were scrambled across generations.

Additionally, research has shown that shorter training tends to favor iconic forms: when learners cannot rely on rote memorization of new form–meaning mappings, they make greater use of iconicity (Kempe et al. 2021). In Kempe et al. (2019), limited training and the absence of explicit communicative demands led to degeneration and reduced distinctiveness, with neither iconicity nor compositional structure emerging. Iconicity and some structural regularities only appeared when participants engaged in a referential communication task. In Kempe et al. (2021), combinatorial structure emerged under both long and short training conditions, but iconicity was specifically strengthened when learning opportunities were limited, with learners exploiting form–meaning resemblance (e.g., shorter signals for small referents).

Finally, some studies have also indicated differences in the strength of iconicity as dependent on the semantic dimension, with stronger effects for size over shape (Erben Johansson, Carr, and Kirby 2021). Comparing the spoken and written domains, iconicity has been found to faster emerge in spoken domains (Vinson et al. 2021).

To sum up, iconicity tends to emerge when learners cannot rely on rote memorization, when meaning (the semantic property of the item) is salient, and when communicative pressure is present. Combinatorial patterns tend to increase with repetition and transmission but are sometimes inhibited by highly iconic or semantically transparent mappings. Learnability typically improves over generations, but whether this improvement is driven by emerging structure or iconicity depends on the design of each study.

Suggested mechanisms underlying the emergence of iconicity

Given what we know from previous work, we suggest three main mechanisms supporting the emergence of iconicity in interactive language learning. Based on the idea that iconicity can facilitate interaction and learning of an artificial language, these mechanisms target different parts of the communication, affecting either the signal being produced by the signal sender,

the ease with which the signal can be mapped onto the intended referent during recognition by the signal receiver, or how the signal sender engages in active strategies to maximize communicative success dependent on past interaction outcomes. The three mechanisms encompass: a *production bias* that biases production toward iconic forms, a *recognition bias* that makes iconic signals easier to identify and remember, and finally, an *adaptive communicative bias* toward making future signals more iconic when communication fails.

Production bias

The *production bias* operates during signal production. In an interaction game, signal senders retrieve a stored signal associated with a meaning and produce a phonetic realization of that signal. Here, production is assumed to be automatically biased toward producing iconic forms, and that the strength of this bias is modulated by the sender's prior experience with the form-meaning mapping. We can ground this possible bias in lexical co-activation. In theories of language production that incorporate continuous, distributed cognitive representations during, retrieving a word will co-activate lexical neighbors that are semantically related (e.g. Buchanan et al. 2025). This semantic similarity might among many other meaning dimensions be physical properties such as size. If a language has a general size-sound-symbolic bias in their lexicon such as English (Winter and Perlman 2021), co-activated lexical neighbors are more likely to exhibit phonetic/phonological properties that iconically relate to its semantic dimensions. Their co-activation can then subtly affect the speech production of the target word (e.g. Goldrick et al. 2016).

The strength of a lexical representation would modulate the dynamics of this co-activation. Low associative strength yields productions that are more variable and more strongly affected by iconicity caused by lexical co-activation, whereas high associative strength yields productions that remain closer to the signal sender's stored forms and are less affected by iconicity. The *production bias* thus represents an automatic, experience-based bias toward iconic targets that weakens as a lexical representation becomes more established. It can thus be thought of as an accidental drift toward iconicity in signal production. Importantly, it makes the assumption that the speaker already has established a rich lexicon that contains iconicity, an assumption that is defensible in light of recent evidence how common iconicity is across languages and semantic dimensions.

Recognition bias

Iconicity has been shown to facilitate lexical processing and learning (Imai and Kita 2014; De Ruiter et al. 2018). Thus, referents associated with iconic signals may be easier to correctly

recognize and learn relative to a neutral signal. This *recognition bias* increases the probability that a referent will be correctly guessed and stored alongside the heard signal.

Adaptive communication bias

In interaction, feedback on communicative success is expected to influence upcoming interactions and signal representations, such that both successful and unsuccessful interactions shape how signals are produced and learned. If iconic strategies arise from communicative pressure, intentionally adapting signals toward more iconic signals should be stronger following unsuccessful trials, when interlocutors struggle to communicate in the emerging language. When communication fails, pressure to improve recognizability shifts signals toward iconic forms, increasing the probability that future productions will be easier for listeners to interpret. When communication succeeds, the produced form is reinforced, stabilizing the emerging convention. The *adaptive communication bias* thus describes an intentional adjustment of signals in response to communicative outcomes, gradually increasing iconicity under uncertainty while stabilizing existing forms when communication succeeds.

Next, we describe how these suggested mechanisms are implemented in a computational model simulating the emergence of iconicity across generations in a reference game between two interlocutors.

Model implementation and underlying assumptions

The model we introduce here simulates the possible emergence of iconicity through iterative learning. Participants learn form-meaning mappings through an interactive game of artificial language learning, played in dyads. The final output of a generation becomes the input for the next generation, simulating language evolution across participant generations. Guided by communicative pressure and incremental learning, the above mechanisms are implemented in the model as parameters possibly causing the emergence of iconicity. The central assumption is that the discussed mechanisms in concert or in isolation lead to the emergence of iconcity over repeated interactions and across generations. Learning is considered gradual and experience-dependent as agents continuously update their internal representations of form-meaning mappings and adjust their signal production based on interaction outcomes (success or failure). Success or failure is determined by the associative strength of the form-meaning mappings, expressed as a probability P of inferring the intended referent. Because learning is considered probabilistic, increases in associative strength do not guarantee perfect communicative success on every trial, but only increase the probability of correct recognition.

The possible biases leading to iconicity are formalized in three sets of parameters: one set of parameters operating on the sender's production of the signal, one operating on the retriever's recognition of the signal, and one on the sender's communicative strategy to modulate the signal based on past experiences. We begin by accounting for the overall structure of the model, before describing how these parameters are implemented.

Overall structure of model

Any simple agent-based model of lexical learning that simulates the emergence of iconicity needs to account for 1) agents and the language to be learnt, 2) how agents produce and perceive form-meaning pairs, 3) how learning incrementally progresses over time and over generations, and 4) the transmission biases affecting this process. As accounted for above, for iconicity to emerge, mechanisms need to exert pressure on agents to favor more iconic signals in production and recognition. The output of interest is whether the signals becomes more iconic over time and whether recognition accuracy increases over time. In this model, iconicity is quantified in a continuous two dimensional signal space as one of the dimensions (vertical dimension).

In the language interaction game simulated here, agents communicate a small number of referents which differ along two semantic dimensions: one semantic dimension has no inherent structure and just represents idiomatic differences and is represented by one axis of the signal space (horizontal axis). Imagine four different-looking alien races are represented by four different consonants (for example b vs. f vs. z vs. k). There is no iconic relationship between the aliens and the consonants. The other semantic dimension, however, represents physical size. Half of the referents are noticeably smaller than the other half of the referents. Physical size is assumed to be perceptually salient, consistent with extensive evidence that magnitude distinctions are readily mapped onto acoustic parameters such as e.g. pitch. Here, the vertical axis of the signal space allows for an iconic mapping. This scenario is akin to the established association between voice pitch and size with small referents being associated with high pitch and large referents being associated with low pitch. Each referent is associated with a unique horizontal parameter value that signals lexical identity (different aliens) and is presented to the speaker alongside an image of the referent. The task for the signal receiver is to identify the intended referent corresponding to the signal produced by the signal sender. The probability of correctly recognizing the referent on each trial reflects the associative strength between signal form and referent–how well the receiver can guess the correct image from the signal.

Referent identity is uniquely cued by the horizontal signal dimension and is always unambiguous to the signal sender. Imagine the signal sender always sees an orthographic representation of the signal that unambiguously represents the horizontal signal dimension alongside the referent image. All the signal receiver needs to do is to remember that the heard signal represents a particular referent.

Before interacting with each other, both agents go through a training phase. During training, they see referents alongside auditory recordings of the signal. All acoustic signals have a neutral vertical signal specification, i.e. neither iconic nor anti-iconic. Imagine a neutral pitch that does not iconically relate to either small or large referents. Since this signal dimension is not orthographically represented, it can be modulated by the signal sender. It does not interfere with the vertical signal dimension. Thus, iconicity does not determine referent identity (different aliens) but can instead indirectly affect recognition within a lexical category. The vertical signal dimension is an additional source of information for recognition of the one semantic aspect of the referent, though it is not strictly required for recognition.

This mirrors a scenario in which meaning is communicated within a conventionalized signal system that is fixed but exhibits additional signal channels that are unused, enabling the communication of other meaning dimensions. It is reminiscent of modern languages in which for example certain acoustic dimensions tightly connected to segmental structure communicate lexical meaning, whereas other aspects of speech such as suprasegmental features (e.g., duration, pitch, intensity) can signal additional meaning dimensions, including iconicity.

The population of each generation consists of one dyad; agent A and agent B. They play an interactive language learning game over 10 rounds. Each round consists of a number of trials; one trial per referent in the language to be learnt. The agents alternate roles as signal receivers and signal senders. Each trial is initialized by randomly assigning agent A and agent B to either role. The sender is given a random referent (different alien that can be either small or large) and the stored signal for that referent is retrieved, including the vertical signal dimension (the consonant context). The sender then produces a signal, and the receiver attempts to recognize the intended referent. The recognition is evaluated as either correct or incorrect, and both agents update their associative strengths about the signal-to-referent mapping, continuously becoming more and more certain about the referent names over trials. Finally, agents update their stored signal representation depending on the interactional success, and a new trial starts. The signal information for each referent in the final round is carried over as signal input to the next generation. This way, iconicity can accumulate over time so that subsequent generations may begin with more iconic signals.

Implementation of suggested mechanisms

The suggested mechanisms are implemented at different stages of the interaction round accounted for above. Figure 1 visualizes a single trial and highlights at what stages the mechanisms are assumed to operate.

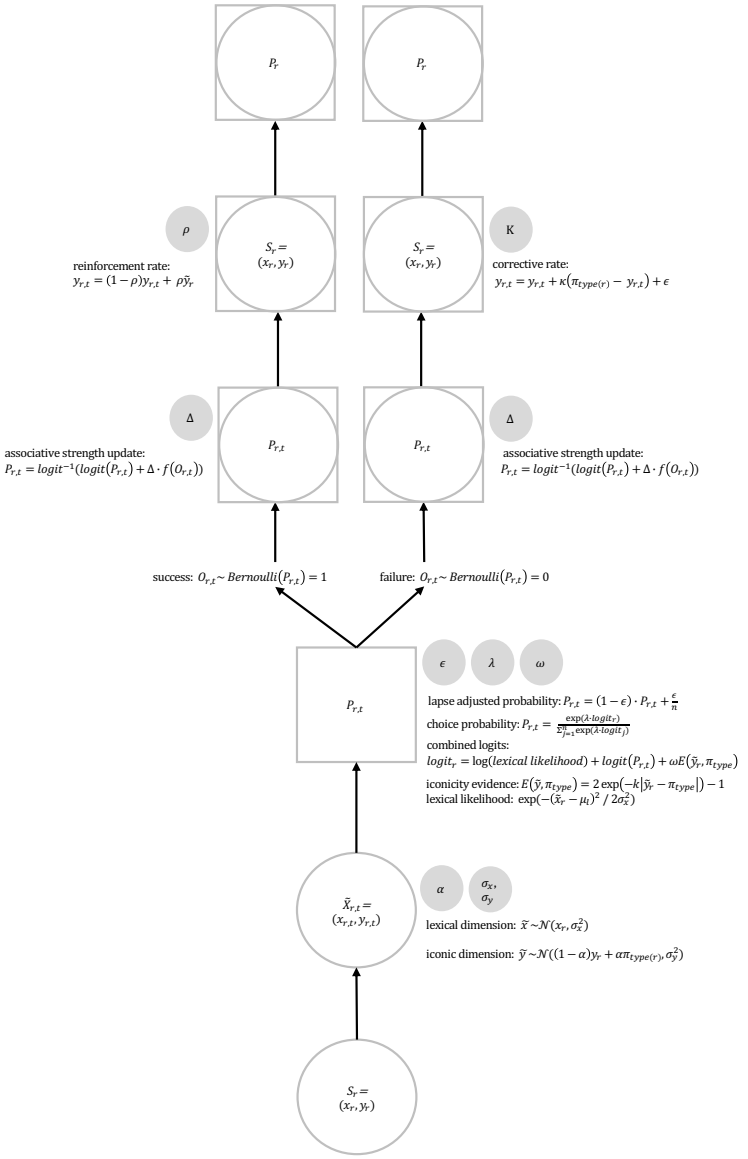


Figure 1: Graphic representation of the iconicity model. The figure shows one trial ($_t$) of the experiment where agents are taking turn as speaker (circles) and listener (squares); from producing a signal ($ ilde{X}{r,t} = [x_{r,t}, y_{r,t}]$), to referent recognition and update of associative strength ($P{r,t}$), and finally updat = [x_{r}, y_{r}]$) before proceeding to the next trial. The three mechanisms are implemented at different steps of the trial: the general articulatory bias at signal production, the interpretation bias at recognition, and the adaptive communication bias at signal update. Grey circles represent mechanisms and their parameters, callout text show formulas.

Producing a signal

Beginning at the bottom of Figure 1, the signal sender is given a referent ($_r$) at random and retrieves the stored signal values, $S_r = (x_r, y_r)$, for the referent. Each referent is associated with 4 pieces of information: a lexeme, a referent identity (represented by a unique image), a semantic property (size: small or large), and a signal value along the iconic dimension. The image uniquely identifies the referent and simultaneously conveys its semantic property (size).

Signals are represented in a two-dimensional space $[x, y]$, but the two dimensions serve distinct functional roles. The x-dimension encodes lexical identity: it is a phonologically distinctive dimension that differentiates lexical items. Each lexeme occupies a fixed prototype location along this dimension The y-dimension encodes iconicity: it is a continuous acoustic dimension ranging from $[0, 1]$ that align with semantic size. Small referents are associated with a prototype at $\pi_{small} = 0$, and large referents with a prototype at $\pi_{large} = 1$. This dimension can be considered a simplified abstraction over multiple correlated acoustic dimensions (e.g., vowel quality, duration, pitch) that jointly can encode size-related meaning. The dimension does not correspond to specific acoustic parameters but captures the idea that signals vary continuously along multi-dimensional perceptual dimensions relevant for iconic mapping. These prototypes represent perceptually transparent signal–meaning correspondences and can be thought of e.g. a formant vowel space in which high front vowels are correlated with small referents and low back vowels are correlated with large referents.

The initial signals at the start of the simulation, i.e., at the first round for the first generation, are placed at the center of the space ($y = [.5]$), meaning that signals are initially not iconic. Lexical prototypes along the x-dimension are evenly spaced and fixed throughout the simulation. For all subsequent generations, the y-values are carried over from the final round of the previous generation.

When producing a signal, $\tilde{X}_{r,t} = (x_{r,t}, y_{r,t})$, the sender samples an x-value from a narrow Gaussian distribution centered on the lexeme's fixed prototype (μ), with phonetic Gaussian noise, $\sigma_x = .01$, to model production variability (including e.g., motor noise). Thus, lexical identity is reliably preserved.

The produced y-value is sampled from a normal distribution whose mean is the stored y-value, and the standard deviation, σ_y, is dependent on the sender's associative strength, $P_{r,t}$—higher previous associative strength reduces the variability along the y-dimension, while lower associative strength increases the variability. This means that the sender's uncertainty about the form-meaning mappings leads to increased variability in the produced signal, which makes it more likely for the signal to drift into iconicity space. As the sender becomes increasingly certain about the form-meaning mappings, the variability along the y-dimension decreases, presumably reducing the risk of walking out of iconicity space again. The baseline standard deviation along y is set to $\sigma_y = 0.022$, which is equivalent to an approximately 10% chance

of wandering into iconicity space when the y signal is at 0.5. Signals are restricted within the set limits ([0,1] in each dimension), so if modulation of the signal results in values outside of its bound, the model truncates the signal to its boundaries.

At generational overturn, the initial learning state of the agents at the first trial is on average a probability of .33. This probability is established by drawing random values from a Beta distribution with a $\alpha = 2$ and $\beta = 4$. As the interaction proceeds, the associative strengths of each agent are carried over across trials.

```
produce_signal <- function(stored_y, lex_prototypes, size_prototypes, speaker_guess, k
    x <- rnorm(1, mean = lex_prototypes, sd = drift_sd_x)
    y <- rnorm(1, mean = stored_y,
               sd = drift_sd_y * (1 + k_production * (0.5 - speaker_guess)))
    c(x, clamp01(y))
  }
```

Recognizing the signal

Moving up in Figure 1, the signal is then presented to the receiver that identifies the intended referent through a combination of previous associative strength and signal-driven evidence along the y-dimension. The resulting recognition probability represents the strength of the form-meaning mapping, where the signal's representation of the semantic prototype is weighted against the receiver's initial guess in log-odds space. Higher previous associative strength generate higher probability of success on subsequent trials, thus reinforcing existing mappings and accelerating learning trajectories.

Iconicity influences recognition through the *interpretation bias* acting on the y-dimension. For each referent, the model computes *signal evidence* as a three-tiered distance between the produced y-value and that referent's prototype. Signals with a moderate distance from the prototype—between 0.2 and 0.8—are considered iconically uninformative, returning 0 evidence. Signals close to their iconic prototype (< 0.2 distance for small referents, > 0.8 distance for large referents) generate positive evidence, scaled exponentially and normalized to a $[0, 1]$ range. Signals far away from the prototype are considered anti-iconic and thus generate negative evidence (> 0.8 distance for small referents, < 0.2 distance for large referents). The sensitivity parameter (k) determines the curvature of this evidence; higher values of k generate a steeper distinction between prototypical 'matches' and the neutral zone, representing a receiver with stricter categorical boundaries.

This evidence is scaled by the *iconicity parameter* (ω), which modulates how strongly proximity to prototypes affects inference. The iconicity parameter is set to 0.043 which corresponds to a 1% absolute increase for an associative strength of 0.5 for a perfectly iconic signal. Together,

these parameters determine the magnitude and sharpness of the iconicity effect on recognition, making form-meaning pairs easier or more difficult to perceive, encode, and retrieve, depending on their spatial proximity to iconic prototypes. Iconicity therefore favors signals that are more closely located to their prototypical target in the two-dimensional space, and punishes signals that are anti-iconic. Figure 2 illustrates signal evidence relative to the signal space, with higher ease for more prototypical values.

```
signal_evidence <- function(produced_y, target, k_perception = 2) {
    dist <- abs(produced_y - target)
    if (dist >= 0.2 & dist <= 0.8) {
      evidence <- 0
    } else {
      edge_dist <- ifelse(dist < 0.2, dist - 0.8, 0.2 - dist)
      magnitude <- exp(-k_perception * edge_dist)
      min_magnitude <- exp(-k_perception * -0.6)
      max_magnitude <- exp(-k_perception * -0.8)
      magnitude_norm <- (magnitude - min_magnitude) / (max_magnitude - min_magnitude)
      evidence <- ifelse(dist > 0.8, -magnitude_norm, magnitude_norm)
    }
    return(evidence)
  }
```

Prior associative strength and iconicity-weighted signal evidence are combined additively in log-odds space and converted back into probability space.

```
listener_guess_probability <- function(listener_guess, produced_y, type, size_prototyp
    icon_ev <- signal_evidence(produced_y, size_prototypes)
    logits <- qlogis(clamp02(listener_guess + (iconicity_weight * icon_ev)))
    probs <- plogis(logits)
    return(list(probs = probs, evidence = icon_ev))
  }

recognition <- listener_guess_probability(
  listener_guess[ref_id],
  produced_y,
  referents_info$type[ref_id],
  referents_info$size_prototypes[ref_id])

prob <- recognition$probs
```

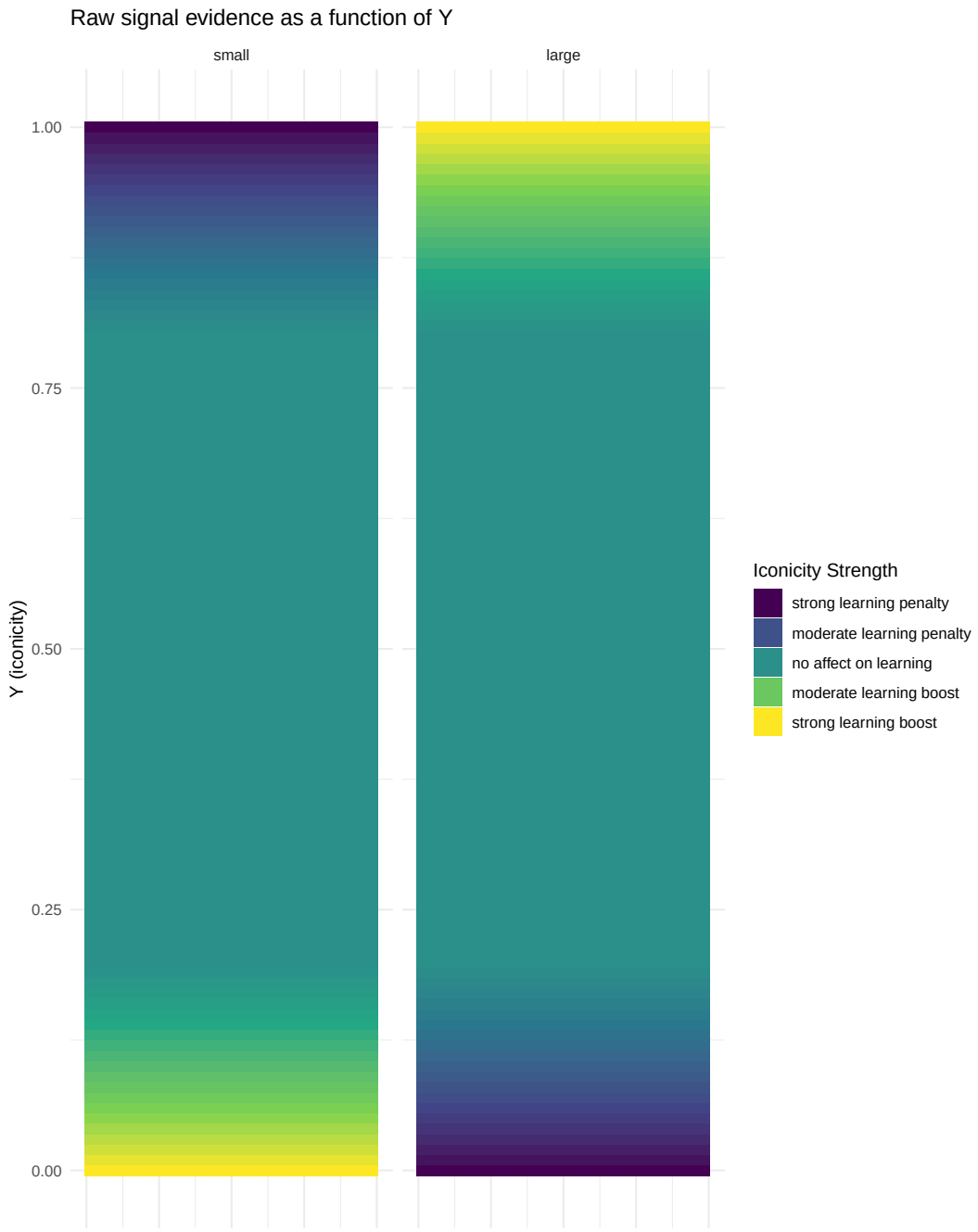


Figure 2: Illustration of signal evidence in the two-dimensional signal space, for small and large referents, for when $k = 5$.

The final outcome of the trial, success or failure, is implemented as random sampling from a binomial distribution with probability = current associative strength; $O_{r,t} \in \{0, 1\}$. A value of 1 indicates that the referent was correctly recognized, while 0 indicates an incorrect guess.

After calculating success or failure, learning is reinforced for both sender and receiver by implementing the *associative learning bias* mechanism. This is a learning mechanism and not an iconicity mechanism. It is parameterized as applying a baseline *learning strength* (Δ) to the sender's and receiver's associative strengths. Successful trials receive a larger update (scaled by 7.5), while successful trials receive smaller update (scaled by 1), reflecting incremental strengthening of form-meaning mappings across interactions, with stronger learning on successful interactions. The learning strength parameter is set to 0.015 corresponding to 1% absolute increase for an associative strength of 0.5 on failed interactions, and a 9% absolute increase on successful interactions.

```
success <- rbinom(1 ,1, prob)

update_logit <- function(x, learning_strength, success, success_scale = 7.5, failure_s
    delta <- learning_strength * ifelse(success == 1, success_scale, failure_scale)
    plogis(qlogis(clamp02(x)) + delta)
  }
```

Updating the signal

After the agents have received feedback on the interaction, the stored y-representation is updated depending on interaction outcome, $S_r = (x_r, y_r)$, Figure @ref(fig:overview-figure). When communication fails, the stored y-representation does not update. When communication succeeds, both the sender and the receiver retrieve the signal for the referent and the stored y-value is updated by a weighted average of the previous y-value and the sender's produced y-value, $S_{y,r} = (y_{produced} + y_{stored})/2$. This implies an experience-based learning mechanisms in which successful interactions gradually reshape representations.

```
reinforce_Y <- function(stored_y, produced_y) {
  ((1 - reinforcement_rate) * stored_y + reinforcement_rate * produced_y)
}

correct_Y <- function(stored_y, referent_type) {
  target_y <- size_prototypes[referent_type]
  clamp01(
    stored_y + corrective_rate * (target_y - stored_y) +
      rnorm(1, 0, drift_sd_y/2))
```

```
}

if(success==1) {
  stored_y[r] <- reinforce_Y(stored_y[r], y_prod)
} else {
  stored_y[r] <- correct_Y(stored_y[r], ref$type)
}
```

Finally, the agents' associative strengths (P_r) are updated based on their current probabilities, before the output is stored and the agents proceed to the next trial. Crucially, lexical categorization is primarily driven by the x-dimension, while the y-dimension provides gradient, iconic support. Iconicity is therefore not required for successful lexical retrieval but can facilitate recognition and shape signal evolution over time. Through repeated interaction, associative learning strengthens form-meaning mappings, while adaptive updating of the y-dimension allows iconic structure to emerge dynamically from communicative pressures. Once iconicity has influenced signal production, the probability of successful communication increases, and the probability of signals drifting out of iconicity decreases. If iconic signals and interactional success accumulate incrementally over interaction rounds, iconic signals are immediately learnable for the subsequent generation as form-meaning mappings are carried over from the previous to the next generation.

Next, we compare the different mechanisms and assess their relative importance for the emergence of iconicity.

Comparison of mechanisms and their relative importance for the emergence of iconicity

For the comparison presented below, we ran 1000 simulations of the model. Each simulation had 1 generations, each generation had 300 interaction rounds with 4 trials each, for a total of 3600000 trials. Simulations were run with fixed random seeds for reproducibility. The number of unique referents in the guessing game was set to 4; 2 small referents and 2 large referents.

The simulations used the following parameter settings: learning strength (.005), iconicity weight (.2), general articulatory bias (.15), reinforcement rate (.05), and corrective rate (.03). Because parameters operate on different computational scales (log-odds space, weighted average, and linear updates), they are not directly comparable in magnitude; however, all were set to relatively small values to produce gradual dynamics where no single bias mechanism should have disproportional influence.

We generated 6 types of models. The main model included all of the above described parameters set at their intialization values, as described above. The remaining 5 models were generated by switching off one of the parameters at a time by setting their value to 0.

Figure 3 illustrates the emergence of iconicity over trials for each of these 6 models. All models display similar patterns of iconicity emergence over time, besides the model for which corrective rate is switched off. This highlights the importance of pushing produced signals on failed interactions toward more prototypical values. Iconicity might thus not be a by-product of communicative success, but requires specific signal-space dynamics. Whereas most of the models have a rather steep increase in iconicity that then levels out, the model where no learning strength is added to the updated speaker and listener guess rates reaches higher iconicity but with a slower increase.

```
# plot
d.iconicity |>
  ggplot(aes(x = total_round, y = evidence, group = simulation,
             color = evidence)) +
  geom_path(size = 0.5, alpha = 0.08,
            color = "black") +
  geom_point(data = d.expressive.agents,
             aes(x = total_round, y = evidence),
             color = "grey",
             size = 1.5,
             alpha = 0.6,
             inherit.aes = FALSE) +
  geom_path(data = d.iconicity.mean,
            aes(group = 1), size = 2,
            color = "black") +
  geom_path(data = d.iconicity.mean,
            aes(group = 1), size = 1) +
  # Add lines at generational overturn
  geom_vline(xintercept = seq(0, 300, by = 30),
             color = "grey",
             lty = "dotted") +
  scale_color_viridis_c(begin = 0,
                        end = 1)+
  scale_y_continuous(limits = c(-1,1), breaks = seq(-1,1,0.25)) +
  scale_x_continuous(breaks = seq(0, 300, 30)) +
  labs(title = "Iconicity over generations",
       y = "Iconicity\n(above 0 = iconic,\nbelow zero = anti-iconic)", x = "Generation
  theme_minimal()
```

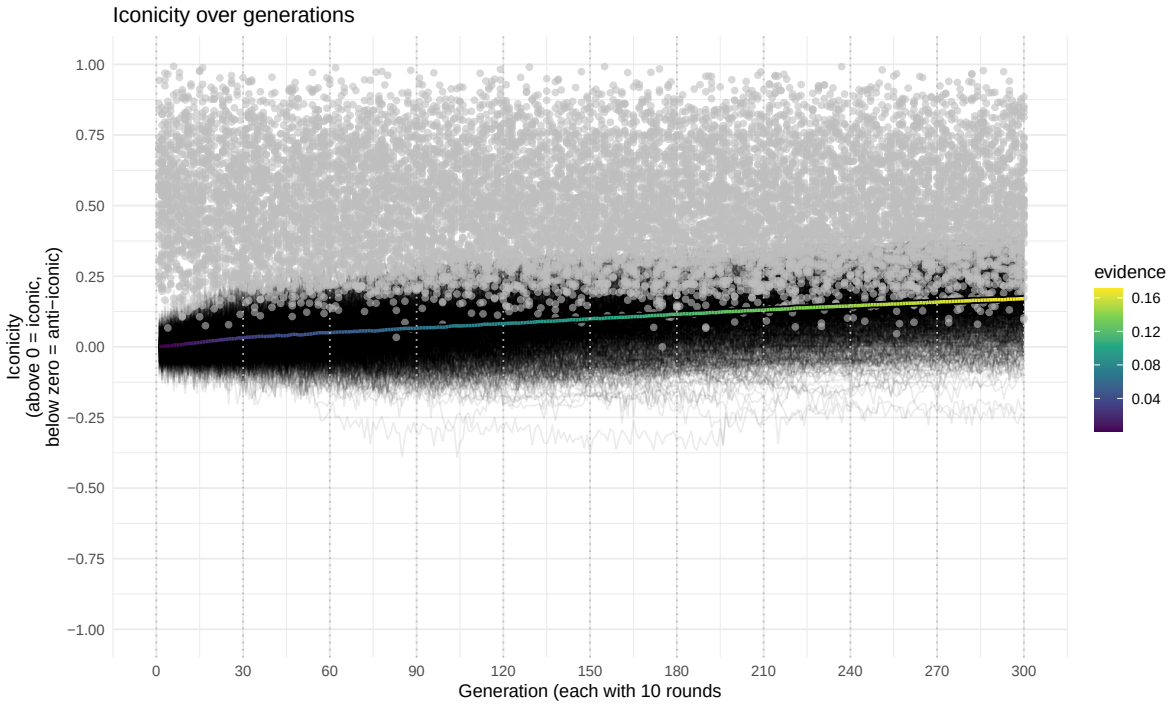
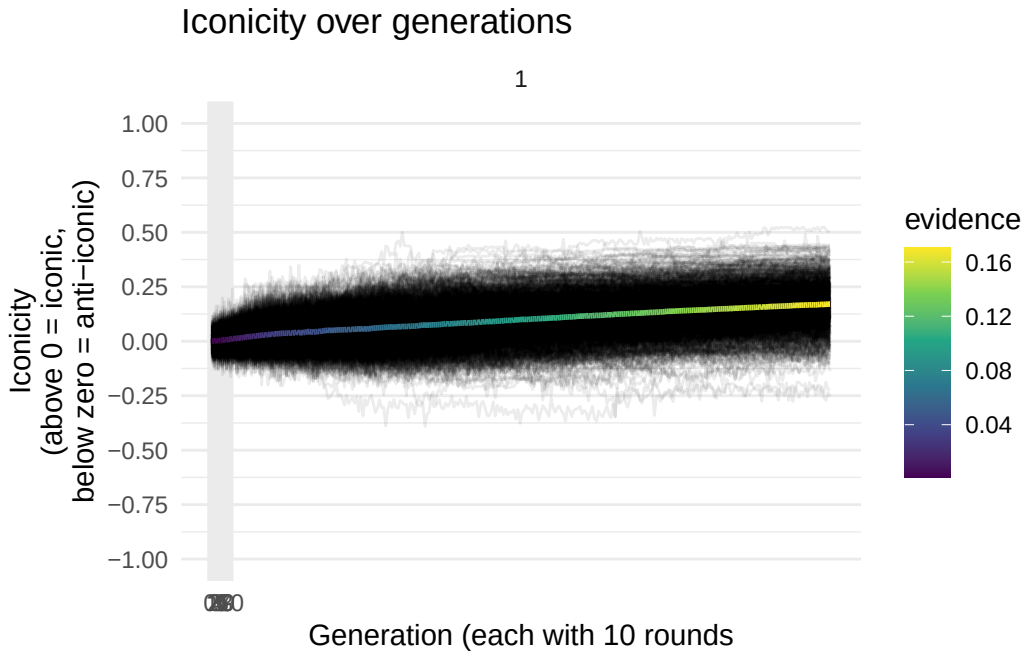
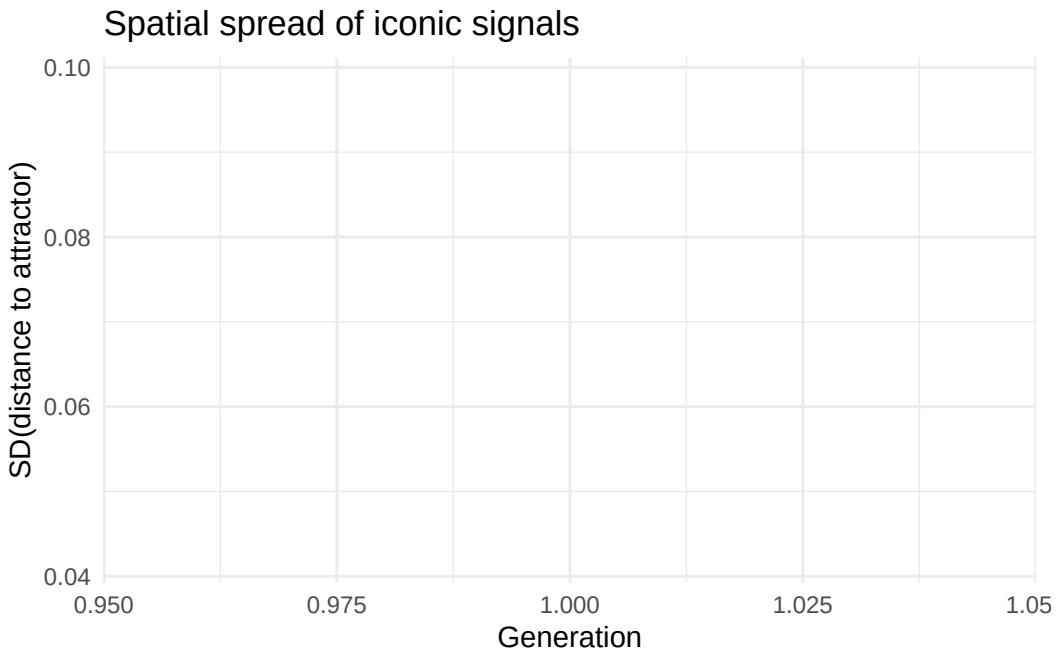


Figure 3: Evolution of iconicity for stored tokens across simulation time. Iconicity is averaged in bins of 100 rounds and showed separately for each parameter setting (facets). Light lines trace individual simulations, the bold line denotes the average across simulations. The dashed horizontal line represents baseline iconicity expected from random signals without structure or learning.

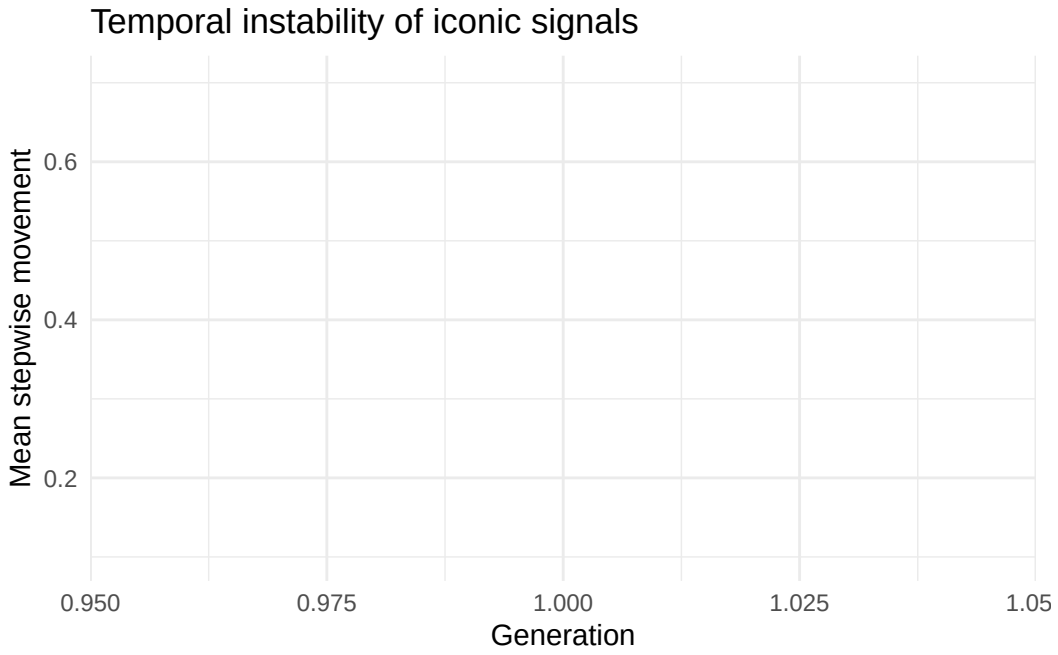



```
# A tibble: 3 x 2
  signal_type proportion
  <chr>            <dbl>
1 anti-iconic      0.156
2 iconic           0.363
3 non-iconic       0.481
```

```
`geom_line()`: Each group consists of only one observation.
i Do you need to adjust the group aesthetic?
`geom_line()`: Each group consists of only one observation.
i Do you need to adjust the group aesthetic?
```




```
`geom_line()`: Each group consists of only one observation.
i Do you need to adjust the group aesthetic?
`geom_line()`: Each group consists of only one observation.
i Do you need to adjust the group aesthetic?
```



We also evaluated the models in terms of the overall success rate across time, that is, the communicative accuracy of agents in interaction. Remember that this refers to the listener's strength in beliefs about the lexeme-referent mappings. The accuracy for each of the 6 models is strikingly similar, except for the model where no learning strength is applied, see **?@fig-accuracy**. This seems to suggest that slower development of iconicity is beneficial for learning?

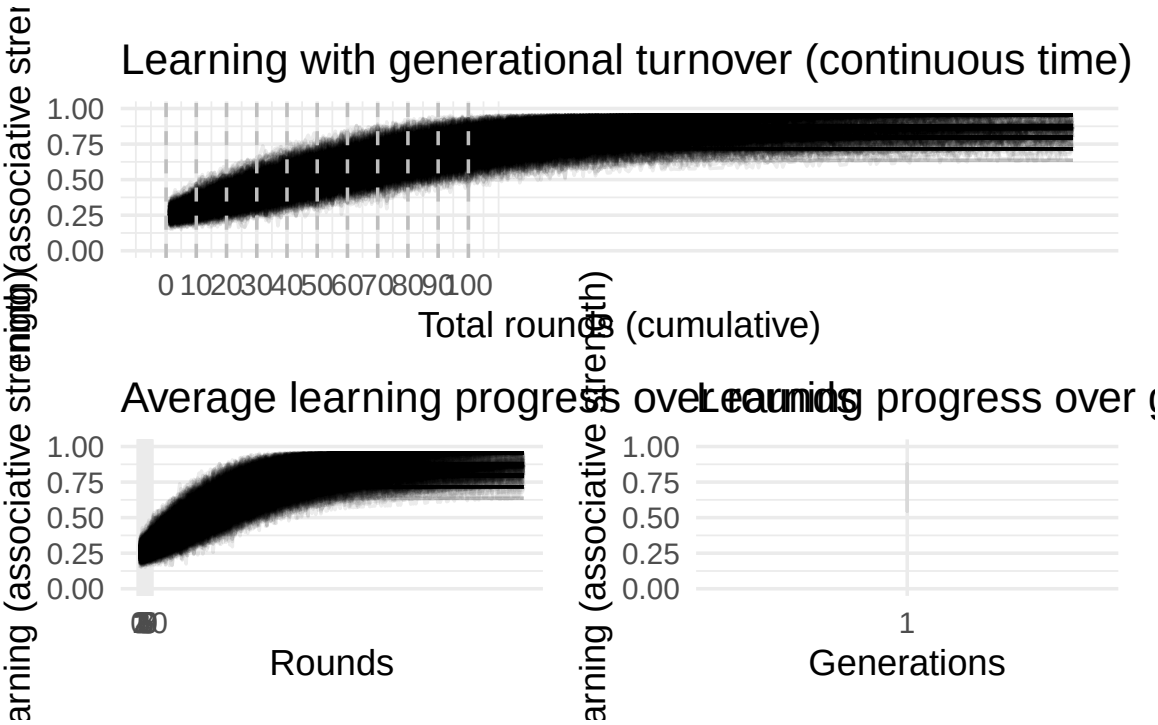


Figure 4: Learning across simulations, black lines correspond to participants' associative strength in beliefs about the form-meaning mappings. The upper panel shows participants' learning progress within each generation, clearly resetting associative strength at generational overturn. Bottom-left panel shows the average learning progress over the 10 rounds, summarised across the 10 generations. Bottom-right panel shows the average learning progress across the 10 generations.

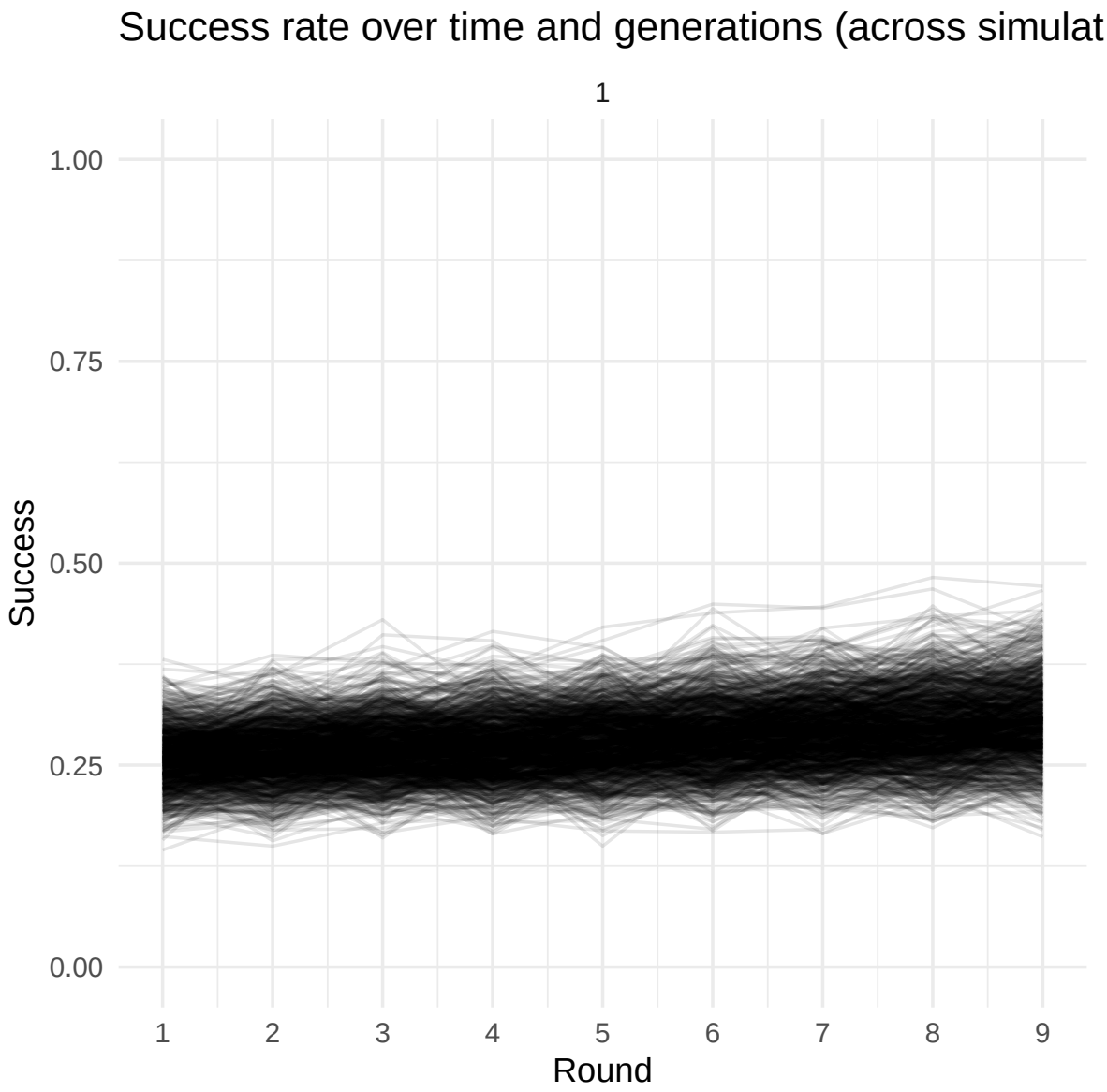


Figure 5: Success rate at the first round for each generation across simulation rounds. Panels correspond to the ten generations.

Figure 6 visualizes the distribution of produced signals across simulations 1, 5 and 10, for each of the 6 model types. The figure seems to suggest that categories separate even for the model where iconicity does not emerge strongly, e.g., the model without corrective rate (third row).

Overall, the results seem to suggest that iconicity emerged reliably except when the corrective

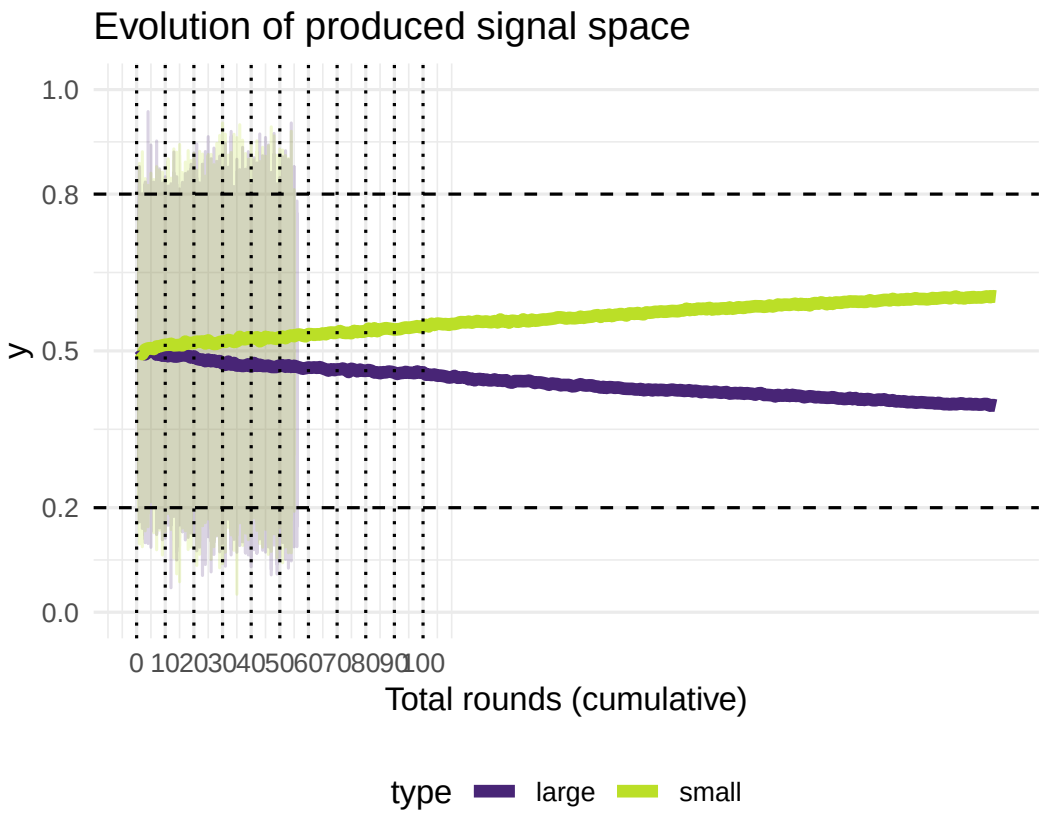


Figure 6: Distribution of produced signals for simulation 1, 5 and 10, along the y-dimension (encoding iconicity). Each point represents a signal token in signal space; colors denote referent type (small vs. large). Columns correspond to simulation replicates and rows to parameter settings.

rate mechanism was disabled. Removing corrective rate eliminated the tendency for signals to move toward their iconic prototypes on failed trials, indicating that this mechanism plays a central role in shaping the signal space. This suggests that iconicity in this model depends on specific signal-space dynamics that bias failed forms toward iconic attractors.

Interestingly, learning strength appeared to influence communicative accuracy without substantially affecting iconicity. When learning strength was removed, agents reached relatively high levels of iconicity but reduced communicative success, indicating that learning strength governs guessing accuracy whereas failure drift governs the emergence of form-meaning structure. This seems to suggest that agents can learn to communicative successfully without producing iconic forms, and that iconic forms can emerge without directly improving recognition accuracy.

Visualizing the learned and produced signal spaces further suggested that category separation can occur even in the absence of iconicity, indicating that lexical categories can form without alignment to iconic prototypes. It might be the case that iconicity emerges from indirect learning advantages rather than from explicit pressure for communicative success—the combination of failure drift, articulatory production bias, interpretative ease, and reinforcement creating a positive feedback loop where failure pushes signals toward iconic prototypes, iconic signals are easier to interpret, successful interpretations reinforce produced forms and reinforcement consolidates iconic patterns in memory?

Grid search over the parameter space

In order to investigate what parameter settings generate the largest increase in iconicity across trials and simulations, we generated a grid search over the five model parameters: articulatory production bias, prototype weighting, reinforcement rate, learning rate, and corrective rate. The grid search explored the following values for each of the parameters.

```
# Set sequences to include baseline in main model
articulatory production bias = c(0, 0.1, 0.15, 0.2)
iconicity weight = c(0, 0.0004, 0.0008, 0.0012, 0.0016)
reinforcement rate = c(0, 0.03, 0.06, 0.09)
corrective rate = c(0, 0.0004, 0.0008, 0.0012, 0.0016)
learning strength = c(0, 0.0004, 0.0008, 0.0012, 0.0016)
```

Similarly to the figures presented above, each panel of **?@fig-grid-search** visualizes iconicity across the parameter space for when one of the parameters is switched off. Replicating previous figures, **?@fig-grid-search**) shows that 1) iconicity struggles to emerge when corrective rate is switched off, 2) learning strength does not seem to support the emergence of iconicity,

and 3) switching off the other three parameters (reinforcement rate, articulatory production bias, prototype weighting) generates highly similar patterns of iconicity, suggesting that these might play less of a role in the model.

For most combinations of parameters, the trade-off between learning strength and iconicity is evident—iconicity is higher for when learning strength = 0, suggesting that the more I learn for free, the less iconic I become (upper two panels, and bottom-right panel, **?@fig-grid-search**). This would suggest that iconicity might not continue to grow in a language, at least not as a learning mechanism, but perhaps as a strategy to make the language more recognizable for the interlocutor.

We also note that under some conditions, iconicity still emerges when corrective rate is switched off (bottom-left panel, **?@fig-grid-search**). This suggests that failure-based drift is not strictly necessary but is normally the dominant pathway (error-driven vs. success-driven iconicity). If this mechanism is not available, iconicity can still emerge by generating a positive feedback-loop of biased production–higher success–reinforcement–more biased learned signals–higher success.

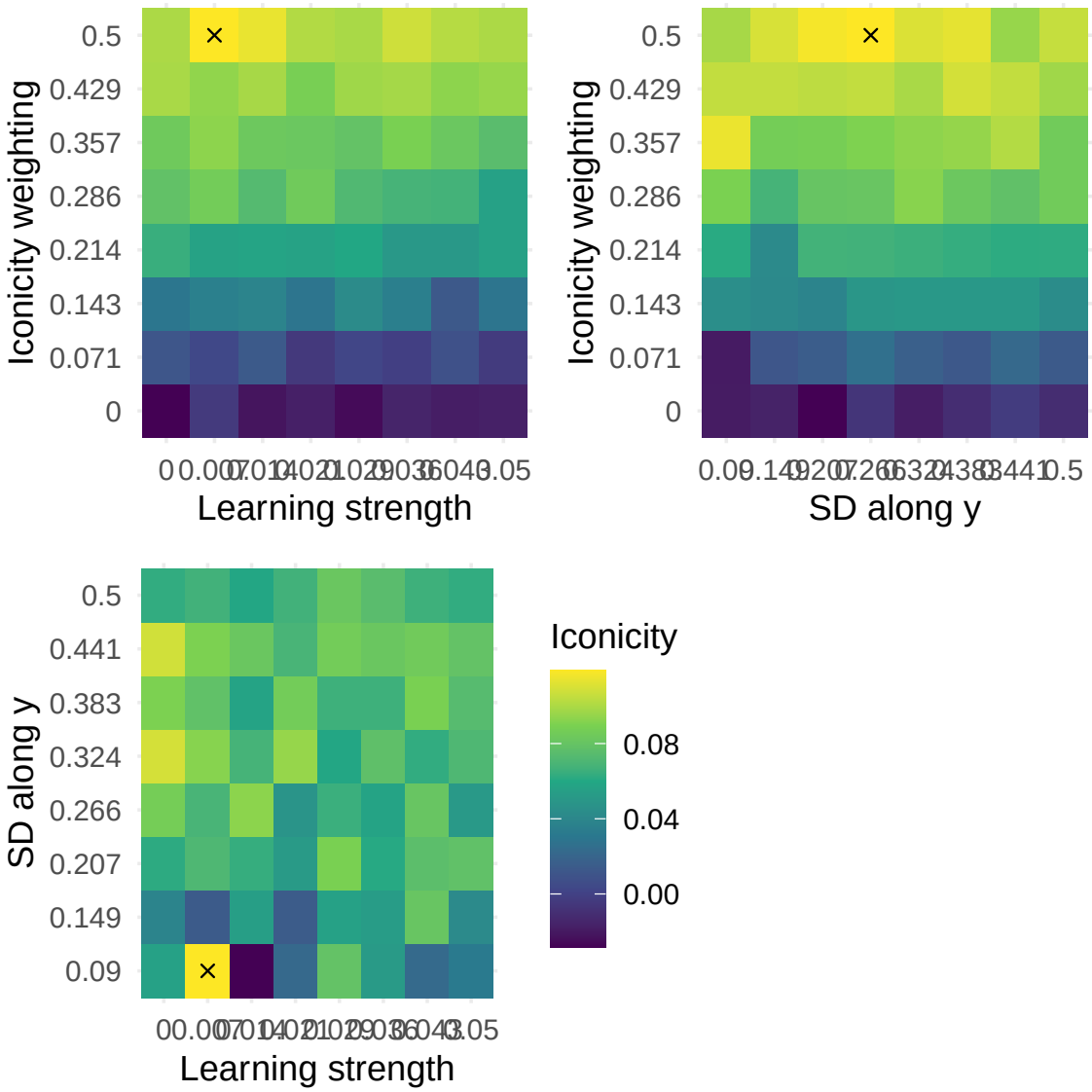


Figure 7: Heatmaps of iconicity across the parameter space for different model configurations. Each panel shows simulations where one parameter is disabled: no reinforcement, no articulatory production bias, no signal drift, no learning, or no prototype weighting. Color indicates mean iconicity; yellow outlines mark the iconicity baseline (~.37) and black crosses indicate the parameter values that generates maximum iconicity.

Open questions

- learning update: the guess probabilities improve over time while corrective rate decreases (no drift on correct trials), this could mean that the signal tends to freeze near its initial state [.5,.5]?
- Why the accuracy-iconicity dissociation?
- Why the emergence of a binary category space even for models where iconicity does not emerge? Could it be that separation is emerging from Bayesian updating + task structure? informative feedback structure?

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